

ON THE SEMI-SUPERVISED SAMPLE COMPLEXITY OF MULTI-OBJECTIVE LEARNING

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MOTIVATION

ML systems must **learn to minimize many risks**:

$$\min_h \mathbf{R}(h) \equiv (R_1(h), \dots, R_K(h)).$$

- *Fairness-aware learning*—decisions simultaneously impact many groups of people
- *Large language models*—correctness, alignment, succinctness, reasoning quality, steerability
- *Self-driving cars*—safety, latency, fuel-efficiency

Question: How much data to learn the Pareto set?

DECISION-MAKING PIPELINE

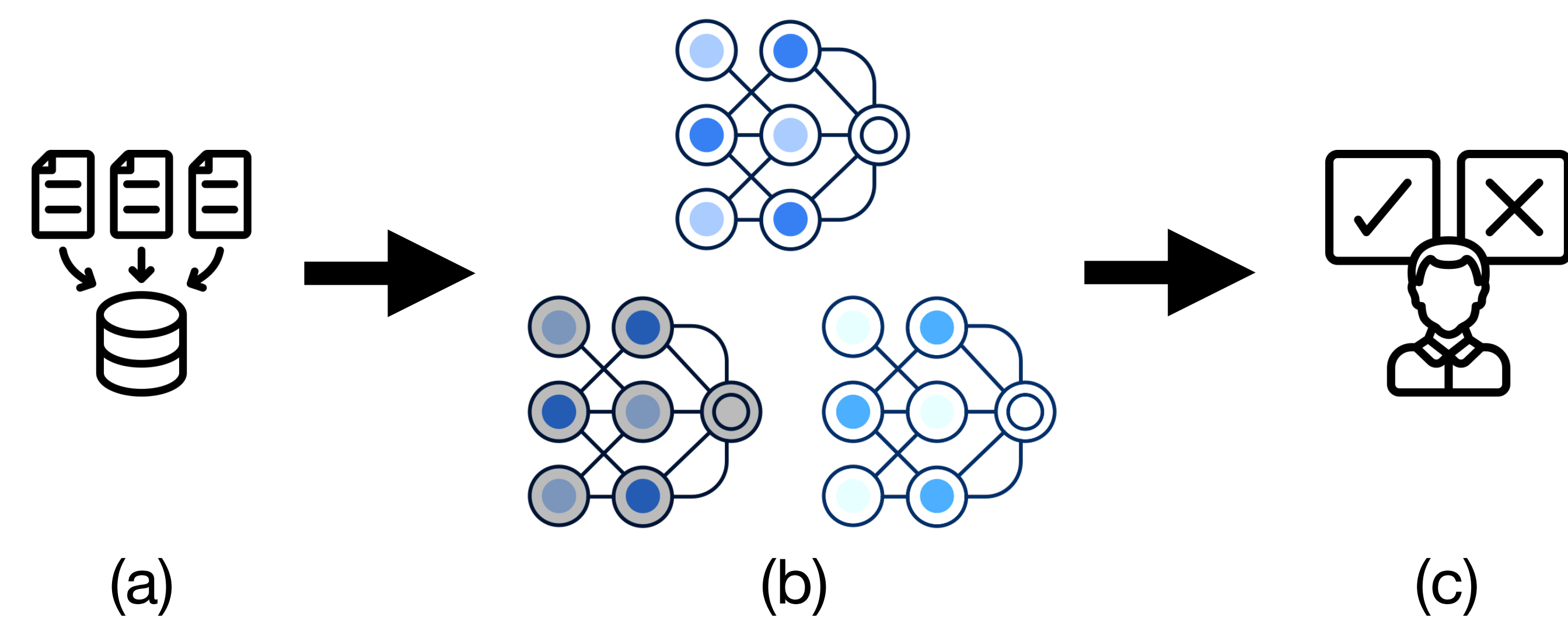


Figure. This work asks (a) how much data is needed to (b) learn all Pareto-optimal models. This allows (c) a downstream decision-maker or human to select the best one to deploy.

LEARNING SETTING

Let $\mathcal{X} \times \mathcal{Y}$ be a data space.

There are K supervised learning tasks:

- Let R_k be a standard supervised learning risk:

$$R_k(h) = \mathbb{E}_{P_k}[\ell_k(Y, h(X))]$$

- P_k is a data distribution
- $\ell_k(y, \hat{y})$ is the loss predicting $\hat{y}$ on y
- f_k^* is the Bayes-optimal model
- $\mathcal{H}_k$ is an individual model class containing f_k^*
- $\mathcal{G}$ is a joint model class containing all $\mathcal{H}_k$

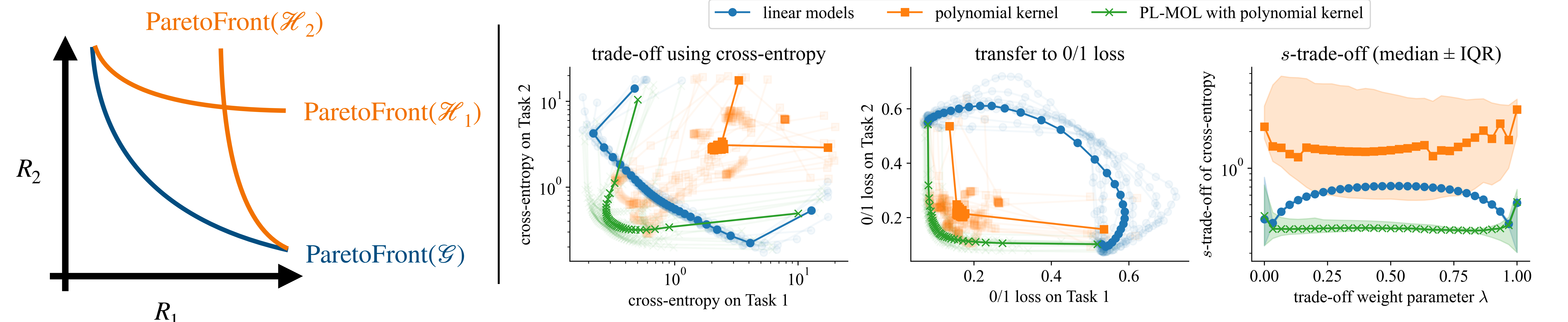


Figure. (Left) Pareto fronts of individual vs. joint model class. (Right) semi-supervision (**PL-MOL**) effectively leverage additional structure.

MINIMIZING RISK VS. TRADE-OFFS

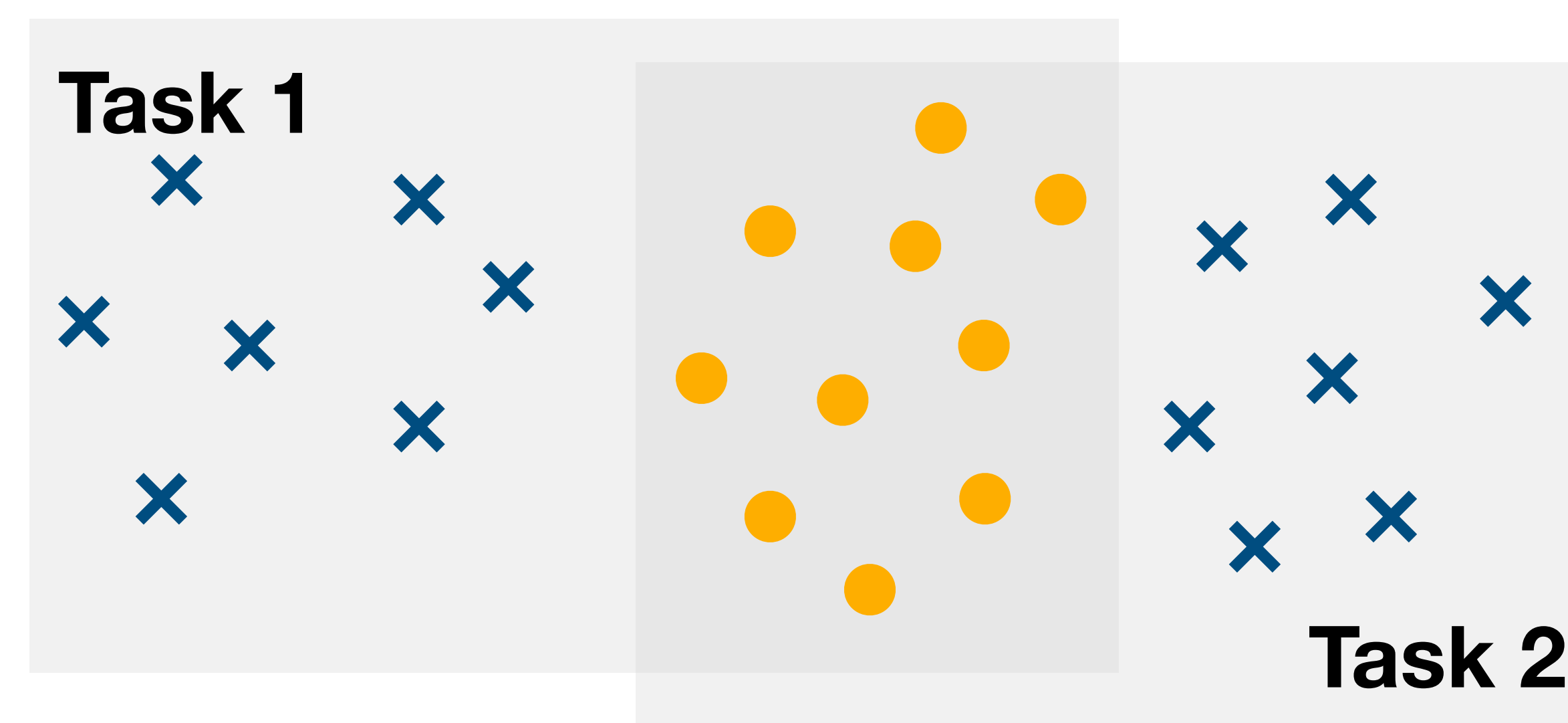


Figure. Good model classes for individual tasks can attain poor trade-offs. Good classes for joint task can be harder to learn.

$\mathcal{S}$ -MULTI-OBJECTIVE LEARNING

Definition. Let $s : \mathbb{R}^K \rightarrow \mathbb{R}$ be a *scalarizer*. Let the *s-trade-off solution* be the solution:

$$g_s = \arg \min_{g \in \mathcal{G}} s(\mathbf{R}(g)).$$

Let $\mathcal{S}$ be a family of scalarizers $s(v_1, \dots, v_K)$

Linear: $\sum \lambda_k v_k$ and Tchebycheff: $\max \lambda_k v_k$

$\mathcal{S}$ -multi-objective learning ($\mathcal{S}$ -MOL). For $\varepsilon, \delta > 0$, return a family of models $\{\hat{g}_s : s \in \mathcal{S}\}$ such that:

$$\Pr\left(\forall s \in \mathcal{S} : s(\mathbf{R}(\hat{g}_s)) - s(\mathbf{R}(g_s)) > \varepsilon\right) < \delta.$$

Question: does knowledge for minimizing individual risk help us make good trade-offs?

A NEGATIVE RESULT

Proposition. Let $\mathcal{G}$ be a VC class and ℓ_k the 0/1-loss. With/without (f_k^*, P_k^X) , solving $\mathcal{S}$ -MOL over $\mathcal{G}$ needs:

$$\Theta(K \cdot \text{VC}(\mathcal{G})/\varepsilon^2) \text{ labeled samples.}$$

The 0/1-loss is too uninformative/unstructured.

APPROACH: PSEUDO-LABELING

- Use labeled data to learn approximate minimizers:

$$\hat{h}_k \approx \arg \min_{h \in \mathcal{H}_k} R_k(h)$$

- Use $\hat{h}_k$ to pseudo-label (PL) a lot of unlabeled data:

$$\{(X_i, \hat{h}_k(X_i)) : X_i \sim P_k\}$$

- Learn $\hat{g}_s$ by minimizing s-trade-off risk on PL data

MAIN RESULTS

Theorem. Let ℓ_k be Bregman losses (e.g. square loss, logistic/cross-entropy loss). The sample complexity is:

$$\text{Label: } \sum \text{VC}(\mathcal{H}_k)/\varepsilon^4; \text{ Unlabeled: } K \cdot \text{VC}(\mathcal{G})/\varepsilon^2.$$

- Amount of labeled data depends only on $\mathcal{H}_k$'s.
- Unlabeled data often much cheaper.
- Fast, localized rates in paper.

Takeaway: Designing loss/feedback with structure is important for efficient multi-objective generalization.